

Grasim – LLM Sales Insights

Case Study

Transforming Sales Analytics with a GenAI-Powered Natural Language Query Platform

Problem Statement	Solution Overview	Application UI	Business Value Achieved
No interactive NL query system on sales data Dashboards underutilised — no analytical insights Voice-to-text capability absent for querying Manual intervention required for every report	3-Layer LLM Pipeline Natural Language Query Input Layer 1 – SQL Generation (LLM) Layer 2 – Insight Generation (LLM) Layer 3 – Visualization (LLM + Python Plotly)	✓ Chatbot interface for NL queries ✓ Auto-generated SQL on BigQuery ✓ Python Plotly charts rendered live ✓ KPI narrative summaries ✓ Unified response payload ✓ Google Cloud API gateway	✓ Decision time cut from 4–5 days to 2–5 minutes ✓ Zero manual report generation ✓ Domain-specific LLM on Vertex AI / Anthropic Claude ✓ Self-service analytics for Sales Head ✓ Scalable on Google Cloud Run GCP Services: Vertex AI · BigQuery · Cloud Run · Cloud Functions · Artifact Registry · IAM

Ganit built an LLM-powered sales analytics platform on Google Cloud, cutting decision time from days to minutes for Grasim's Sales Head

Outcome

Delivered a cloud-native GenAI query platform on Google Cloud for Grasim's Supply Chain and Digital teams — fine-tuned on domain-specific sales data, generating SQL queries, KPI insights, and Python Plotly visualizations in under 5 minutes from a single natural-language question.

Operational Efficiency	Cost Optimization	Scalability & Reliability	Business Growth
Eliminated manual report generation. The Sales Head can now query sales data in plain English and receive structured insights, charts, and SQL results in 2–5 minutes — down from 4–5 working days.	Replaced manual analyst intervention with an automated 3-layer LLM pipeline on Google Cloud Run and Cloud Functions — significantly reducing operational overhead and reporting costs at scale.	Built on Google Cloud with Vertex AI (Anthropic Claude), BigQuery, Artifact Registry, and Cloud Run — supporting concurrent users with auto-scaling, incremental delta refreshes, and governance via IAM.	Enabled real-time, data-driven decisions for Supply Chain and Digital sales teams — with self-service analytics, live visualizations, and domain-tuned LLM understanding of Grasim's product and territory terminology.

No interactive query system and underutilised dashboards blocked real-time sales insights for Grasim's leadership team

Problem Statement

Grasim's Supply Chain and Digital teams relied on static dashboards and manual analyst reports for sales decisions. There was no interactive system to query sales data, no voice-to-text interface, and no mechanism to extract insights without significant manual effort — causing decision delays of 4–5 working days.

Challenge 1

No interactive system to give responses based on sales data — queries required manual analyst intervention.

Challenge 2

Current dashboards weren't being efficiently utilised to gain sales analytical insights for leadership decisions.

Challenge 3

No voice-to-text or natural language interface for asking questions about sales performance or distributor data.

Challenge 4

Sales Head required 4–5 working days to receive insights — incompatible with fast-moving commercial decisions.

Ganit built a multi-layer GenAI platform on Google Cloud — LLM-driven SQL generation, insight summarization & Plotly visualizations from a single NL query

Solution Overview

Robust & Scalable Architecture

- LLM fine-tuned on Grasim domain data via Vertex AI (Anthropic Claude) — maps business terms to BigQuery metadata.
- Layer 1: Autocorrect layer (Rephrasing + NER + Similarity Search) cleans and normalises user query before SQL generation.
- Layer 2: SQL Generator (Phase I + II) produces validated BigQuery SQL from structured user intent.
- Layer 3: Graph Generator (Phase I–III) + Insight Generator produce Plotly charts and KPI summaries from the DataFrame.

Application UI

- API gateway (Cloud Run) serves unified response payload — SQL output, charts, and narrative insights.
- Google Cloud Functions handle preprocessing; Artifact Registry manages container images.
- Cloud Pub/Sub + Cloud Logging for event-driven orchestration and monitoring.

Data Architecture

